

Nuzhat Nuari Khan Rivu

Iowa City, IA 52246

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[Google Scholar](#)

[LinkedIn](#)

EDUCATION

Ph.D. in Applied Mathematical and Computational Sciences

Department of Mathematics, University of Iowa

Aug. 2024–Present

Master of Science in Mathematics

Department of Mathematics, University of Dhaka

Apr. 2023–May 2024

CGPA: 3.81/4.00

Bachelor of Science in Mathematics

Department of Mathematics, University of Dhaka

Jan. 2018–Jan. 2023

CGPA: 3.60/4.00

RESEARCH INTERESTS

- Applied Mathematics
- Data Science
- Mathematical Biology
- Bioinformatics

PUBLICATIONS

- [1] **Nuzhat Nuari Khan Rivu**, Md Kamrujjaman, and Asif Iqbal. “HIV/AIDS suppression in North America: Intervention plans and cost-effectiveness of UNAIDS 90-90-90 and 95-95-95 targets.” *Communications in Nonlinear Science and Numerical Simulation*, Vol. 152, Part B, 109246, 2026. [↗](#)
- [2] **Nuzhat Nuari Khan Rivu**, Md Kamrujjaman, and Shohel Ahmed. “Comparative analysis of stochastic and predictable models in the HIV epidemic across genders.” *International Journal of Biomathematics*, Online Ready, 2025. [↗](#)
- [3] **Nuzhat Nuari Khan**, Shahin Ara Begum, Raheedun Afeef, and Md Kamrujjaman. “Vaccine efficacy of COVID-19 in Bangladesh: Does vaccination prevent the pandemic?” *GANIT: Journal of Bangladesh Mathematical Society*, Vol. 43, No. 1, pp. 45–62, 2023. [↗](#)

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** Aug. 2024–Present
Department of Mathematics, University of Iowa, Iowa City, IA
 - **MATH:0100 Basic Algebra I** Fall 2026
Standalone course
 - **MATH:1560 Engineering Calculus II** Spring 2026
Two discussion sections
 - **MATH:1005 College Algebra** Fall 2025
Standalone course
 - **MATH:1350 Quantitative Reasoning for Business** Spring 2025
Two discussion sections
 - **MATH:1550 Engineering Calculus I** Fall 2024
Two discussion sections

ACADEMIC & PROFESSIONAL EXPERIENCE

Graduate Research Assistant

University Grant Commission (UGC), University of Dhaka

Apr. 2023–May 2024

Mathematics Instructor

Udvash Academic and Admission Care

Jan. 2019–May 2024

• National Science and Technology (NST) Fellowship	<i>2023</i>
Awarded for M.S. thesis research in the Physical Science group.	
• Sumitomo Corporation Scholarship	<i>2020–2024</i>
Awarded based on academic achievement and merit.	
• Women’s Mathematics Olympiad	<i>2019–2022</i>
2022: 1st Place	2021: 2nd Place
2020: 8th Place	2019: 6th Place
• EWU Interuniversity Mathematics Olympiad	<i>2019</i>
4th Place	
• Dhaka Board Scholarship	<i>2017</i>
Talent Pool Scholarship; Merit Position: 60th in Dhaka Board.	

Organizing Volunteer <i>International Day of Mathematics</i>	<i>Mar. 2023</i>
Organizing Secretary, Communication and Media <i>Dhaka University IT Society (DUIITS)</i>	<i>Jan. 2021–Jan. 2023</i>
Membership Coordinator <i>Vigarunnisa Noon Debating Club</i>	<i>Jul. 2015–Mar. 2017</i>

Programming:	MATLAB, Python, Fortran
Mathematical Computing:	Mathematica
Scientific Writing:	L ^A T _E X
Data Analysis:	Microsoft Excel

Junior Certificate in Spanish Language (A1) Institute of Modern Languages, University of Dhaka	<i>Apr. 2022–Mar. 2023</i> 1 st Division
What is Data Science? IBM, Coursera	<i>Aug. 2023</i>
Data Science: Foundations Using R Specialization IBM, Coursera	<i>Jul. 2021</i>
Python for Everybody Specialization IBM, Coursera	<i>Nov. 2020</i>

Dr. Colleen Mitchell Associate Professor Department of Mathematics University of Iowa colleen-mitchell@uiowa.edu	Dr. Laurent Jay Professor Department of Mathematics University of Iowa laurent-jay@uiowa.edu
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